

Recursive Deep-Dictionary Reconciliation

In cloud configuration management and complex data integration tasks, merging hierarchical datasets is a frequent requirement. Unlike a shallow update, a "Deep Merge" must recursively reconcile nested structures, ensuring that sub-dictionaries are merged rather than overwritten, allowing for fine-grained overrides of specific configuration leaf nodes.

Implement a recursive `deep_merge` utility function with type hints that reconciles two multi-level dictionaries. The engine must prioritize values from the "source" dictionary while recursively drilling into nested dictionaries found in both inputs to ensure a lossless integration of complex hierarchical data.

Example 1 Input:

```
MERGE {"a": 1, "b": {"c": 2}} {"b": {"d": 3}, "e": 4}
```

Output:

```
{"a": 1, "b": {"c": 2, "d": 3}, "e": 4}
```

Explanation: The dictionaries share a common key `b`. Instead of overwriting `{"c": 2}` with `{"d": 3}`, the recursive merger integrates the two sub-dictionaries, preserving all nested keys.

Example 2 Input:

```
MERGE {"a": [1]} {"a": [2]}
```

Output:

```
{"a": [2]}
```

Explanation: Since the values associated with key `a` are lists (not dictionaries), the recursive logic performs a standard overwrite, replacing the base list with the source list.

Function Parameter:

- `base (dict)`: The target dictionary to be updated.

- `source (dict)`: The dictionary containing overriding or supplementary data.

Returns

- `dict`: The fully reconciled hierarchical dictionary.

Constraints

- The implementation **must** be recursive for nested dictionary types.
- Non-dictionary types (lists, integers, strings) should follow standard overwrite behavior.
- The original input dictionaries should remain unmodified (return a new object).

Input Format for Custom Testing Input from stdin contains: `MERGE <json_base>`
`<json_source>`.

Sample Case 0 Sample Input

```
MERGE {"x": 10} {"y": 20}
```

Sample Output

```
{"x": 10, "y": 20}
```

Explanation Two simple dictionaries with no overlapping keys are successfully combined into a single flat structure.